Ex. No: 2a

Experiment Title: Shell Script

Aim:

To write a shell script to display a basic calculator performing arithmetic operations.

Procedure:

1. Start the terminal in your Linux environment.
2. Open a text editor and create a file named `arith.sh`.
3. Write a shell script that:
   - Prompts the user to enter two numbers.
   - Uses arithmetic expressions to calculate addition, subtraction, multiplication, division, and modulo.
   - Checks for division by zero using an `if` condition.
4. Save the file and give it execute permission using `chmod +x arith.sh`.
5. Run the script using: `sh arith.sh`
6. Verify the output for different sets of inputs.

Program:

Filename: `arith.sh`

Unset
```
#!/bin/bash

echo "Enter two numbers:"
read a
read b

echo "add $((a + b))"
```

```
echo "sub $((a - b))"
echo "mul $((a * b))"

# Check for division by zero
if [ "$b" -ne 0 ]; then
    echo "div $((a / b))"
    echo "mod $((a % b))"
else
    echo "div: Division by zero error"
    echo "mod: Division by zero error"
fi
```

Sample Input and Output:

```
Unset
[REC@localhost ~]$ sh arith.sh
Enter two numbers:
5
10
add 15
sub -5
mul 50
div 0
mod 5
```

Result:

The shell script was executed successfully. It accepted two numbers and displayed the result of addition, subtraction, multiplication, division, and modulo operations.

Ex. No: 2b

Experiment Title: Shell Script

Aim:

To write a shell script to test whether a given year is a leap year using a conditional statement.

Procedure:

1. Open the terminal and create a new shell script file named `leap.sh`.

2. Write the script to:

    - Prompt the user to enter a year.

    - Use conditional statements to check if the year is divisible by 4 but not by 100, or divisible by 400.

    - Display whether it is a leap year or not.

3. Save the script and make it executable: chmod +x leap.sh
4. Run the script
5. Test the program with various year inputs (e.g., 2000, 1900, 2024).

Program:

Filename: `leap.sh`

Unset

```
#!/bin/bash



echo "Enter a year:"

read year



if [ $((year % 4)) -eq 0 ]; then
```

```
    if [ $((year % 100)) -ne 0 ] || [ $((year % 400)) -eq 0 ];
then
        echo "Leap year"
    else
        echo "Not a leap year"
    fi
else
    echo "Not a leap year"
fi
```

Sample Input and Output:

Unset

```
[REC@localhost ~]$ sh leap.sh
Enter a year:
2012
Leap year
```

Result:

The shell script was executed successfully. It checked the given year and correctly determined whether it is a leap year or not using conditional statements.